The dog–owner dyad in air transport in the cargo hold: the human as the source of stress, the dog as emotional agent, and the mismatch between perceived and observed risk

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Abstract

Objective. To argue that the emotional and physiological consequences of air transport in the cargo hold are not explained by the pressurized compartment itself, but by a coupled dyad in which the human is the source of the stress and the dog, after millennia of domestication, acts as its emotional agent. **Method.** Interdisciplinary technical note integrating official incident data (US DOT, 2025), comparative stress physiology (HPA and sympathoadrenomedullary axes), evidence of human-to-dog emotional transmission and attachment theory, with verification of every reference against its primary identifier (DOI/PMID) and prioritization of Q1 sources. **Results.** Observed physical risk is low (US DOT: 99.99 % incident-free across 117,861 animals in calendar year 2025), revealing a mismatch with owner-perceived risk. The neuroendocrine stress axis is shared by human and dog, a substrate that allows physiology and emotion to be joined. Acute transmission is directional and experimentally measured: human fear chemosignals increase stress behaviours and heart rate in the dog (D'Aniello et al., 2018, Animal Cognition, Q1). To this is added a long-term dyadic coupling, the hair-cortisol synchrony between dog and owner (Sundman et al., 2019, Scientific Reports, Q1), whose directionality the design does not resolve and which this work treats as a testable hypothesis. Separation from the attachment figure and flight stressors act as triggers upon that dyad. Tranquilizing sedation lacks demonstrated efficacy (Bergeron et al., 2002). **Conclusion.** The engine of travel distress is the emotional state of the human, who produces symptoms of their own and, through evolutionary coupling, transmits them to the dog; regulating the human is part of managing the animal. The contribution is the interdisciplinary integration of veterinary physiology and human psychology around a dyadic phenomenon, in an explicitly observational and delimited mode, subordinate to individual veterinary assessment.

Keywords: dog-human dyad, emotional agent, human-to-dog emotional transmission, perceived vs observed risk, HPA axis, cortisol synchrony, chemosignals, air cargo transport, animal welfare

1. Introduction and rationale

The transport of companion animals in the cargo hold of commercial aircraft generates a recurrent concern in owners, usually expressed in two questions: whether the animal is at physical risk during the flight, and how its distress may be reduced. This technical note argues that these two questions belong to different frameworks, and that conflating them explains much of the anxiety surrounding the procedure.

The purpose of this work is to examine whether the pressurized hold constitutes an intrinsic risk to the dog, and to integrate, within a single interdisciplinary framework, four domains that are rarely analysed together: official air-transport incident data, comparative stress physiology, evidence of emotional transmission between human and dog, and attachment theory. The thesis is twofold. First, that the risk perceived by the owner, centred on the pressurized compartment, is misaligned with the risk observed in official records. Second, and of greater reach, that the engine of distress does not reside in the animal in isolation but in the dyad: the owner is the primary source of stress, producing symptoms of their own and, through a coupling forged by millennia of domestication, transmitting them to the dog, which has been evolutionarily selected to read and mirror human emotional states. Within this framework the dog is not merely a patient that becomes stressed: it is an emotional agent of its caregiver, and its distress during travel is largely derived from that of the human.

2. Observed risk: the actual magnitude of the adverse event

The United States Department of Transportation (US DOT) requires commercial airlines to report every death, injury or loss of a transported animal, and publishes the totals periodically. According to the annual animal report for calendar year 2025, filed under 14 CFR Part 235 and published in the February 2026 edition, 117,861 animals were transported on US airlines, with 3 reported deaths; 99.99 % of animals arrived without any incident. This is the consolidated annual figure, not a partial one. The compartment designated for live animals is a pressurized and climate-controlled space that broadly shares the pressurization system of the passenger cabin.

These data delimit the observed physical risk as low in population terms. The absence of a clinical incident, however, is not equivalent to the absence of a stress response, and this distinction structures the remainder of the review.

3. The dog–caregiver bond as an attachment system

The clinical relevance of separation follows from this framework: transport in the hold removes the figure that regulates the animal's emotional response precisely at the moment of maximal exposure to novel and potentially threatening stimuli.

4. Pathophysiology of separation during transport

4.1 Neuroendocrine activation

The stress response is organised along two complementary axes. The sympathoadrenomedullary (SAM) axis responds immediately through catecholamine release, raising heart rate and blood pressure. The hypothalamic–pituitary–adrenal (HPA) axis releases cortisol, which sustains the response and participates in its resolution. Bergeron et al. (2002) provide the only series with physiological measurements in dogs during an actual commercial flight: in beagles, salivary cortisol rose significantly and heart rate reached its greatest increases during loading and unloading rather than during stable flight. Leadon and Mullins (1991) had previously documented the stress response in real flight. In ground transport, Herbel et al. (2020) demonstrated cortisol elevation at every episode without habituation across repeated exposures, and Beerda et al. (1997, 1998) characterised cortisol, heart-rate and behavioural responses to acute stressors.

4.2 Cardiovascular correlate

Sustained sympathetic activation has measurable consequences. Wormald et al. (2017) documented reduced heart-rate variability in dogs with anxiety-related problems, indicative of lower parasympathetic tone and unexplained by behaviour or cortisol. Soghyani et al. (2025) found an association between anxiety and echocardiographic parameters of cardiac function. In animals without cardiac disease these responses are transient; in the presence of cardiac disease they operate upon an already reduced functional reserve.

4.3 Respiratory correlate

Panting is simultaneously a sign of stress activation and the principal thermoregulatory mechanism of the dog. Sustained hyperventilation may induce respiratory alkalosis; Albers et al. (1975) described this phenomenon, albeit in a model of thermal stress, so that its extrapolation to panting of emotional origin requires caution. Clinical relevance concentrates on the interaction with moderate cabin hypobaria and with brachycephalic conformation.

5. The human as origin of the stress and the dog as emotional agent

This is the core of the argument. The owner's stress response is not independent of the animal's: it is its origin. Two features make this possible. The first is a shared physiological substrate : the hypothalamic–pituitary–adrenal axis and the sympathoadrenomedullary system operate in the dog with the same architecture as in the human, so that emotion and physiology are not separate domains but a single stress-response machinery common to both species. The second is an evolutionary coupling : domestication selected in the dog an exceptional sensitivity to human emotional states, to the point of making it an emotional agent of its caregiver.

It is worth specifying what each piece of evidence establishes, because they do not all prove the same thing. The human-to-dog direction is experimentally measured in the acute mechanism : D'Aniello et al. (2018), in Animal Cognition , exposed dogs to body chemosignals collected from human donors under fear , happiness or a control condition, and recorded increased stress behaviours and heart rate in the fear condition. Because the experimental design imposes the exposure, the causal arrow is constrained: the human emotional state is transferred to the dog through a channel the animal does not choose.

Sundman et al. (2019), in Scientific Reports , document something different and complementary: hair cortisol levels of dog and owner synchronise over the long term across 58 dyads, with stronger coupling in females. This is a finding of longitudinal co-coupling whose directionality the design does not distinguish — synchrony is compatible with the human driving the dog, with the reverse, or with both — and Section 8 proposes precisely to test it in its acute form. Konok et al. (2015) add that the owner's attachment style and personality predict the severity of the dog's separation-related disorder: an association in which the human variable precedes the animal's, congruent with the proposed direction although observational in nature.

The joint reading is therefore more precise than common intuition: there exists an acute human-to-dog transmission pathway demonstrated experimentally, and there exists a long-term dyadic coupling whose initiative remains unresolved. The model proposed here rests on the former and treats the latter as a testable hypothesis rather than an established fact.

The consequence for transport is direct. An owner arriving at the cargo counter with their own stress response activated — unresolved uncertainty about the animal's welfare, anticipation, guilt — is not an external observer of their dog's distress: they are, in part, its cause. Regulating the caregiver's emotional state ceases to be a courtesy and becomes an intervention upon the patient.

6. Risk stratification

Certain profiles present reduced reserve for coping with the sum of separation and environmental stressors, which gives individual veterinary assessment decisive weight. In brachycephalic breeds, increased upper-airway resistance limits ventilatory compensation and heat dissipation by panting under stress (Ladlow et al., 2018; Packer et al., 2015); obstructive airway syndrome appears as the leading cause of mortality in some of these breeds in clinical settings (Wong et al., 2025). In cardiac disease, sympathetic activation and reduced heart-rate variability operate upon a limited reserve. In pre-existing separation anxiety — with population prevalence between 14 % and more than 50 % — transport activates an already established disorder (Ogata, 2016; Flannigan and Dodman, 2001; Lenkei et al., 2021). Geriatric animals and very young puppies present, respectively, reduced reserve or immature regulation.

7. Evidence-based management

Supported interventions aim to restore a secure base and attenuate the stress response without compromising the animal's physiology. Systematic desensitisation and counter-conditioning to the travel crate during the preceding weeks turn the container into a portable secure base; Mariti et al. (2012) documented that dogs habituated to transport from early life show a substantially lower incidence of travel-related problems. Dog-appeasing pheromone improves the behavioural profiles of travel-related problems (Gandia Estellés and Mills, 2006). Frank et al. (2006), in a placebo-controlled double-blind trial, demonstrated that clomipramine reduces post-transport cortisol without producing motor sedation, evidence that anxiolysis is dissociable from tranquillisation.

Routine tranquillising sedation is not supported. In the real-flight series of Bergeron et al. (2002), acepromazine did not modify any physiological or behavioural stress variable. In addition, sedation may depress ventilatory and cardiovascular responses and compromise thermoregulation and postural stability, with risk increased by cabin altitude and particular relevance in brachycephalic breeds; consequently, veterinary associations and air-transport regulations advise against its routine use. The choice and indication of any pharmacological intervention rest with the attending veterinarian and with airline regulation.

8. A dyadic model of transport distress and its testable predictions

The literature has almost invariably treated the dog's transport distress as a problem localised in two places: the animal — its physiology, its temperament — and the environment — the hold and its physical stressors. This work proposes to relocate it. If the evidence of a shared neuroendocrine substrate and of directional emotional transmission is correct, transport distress is a phenomenon neither of the animal nor of the compartment, but of the dyad , with the owner as source and the dog as emotional agent. We term this reformulation the coupled-dyad model .

A model is useful only if it risks predictions that could prove false. The coupled-dyad model generates at least five, all testable with current technology and without instrumenting a cargo hold:

- Owner anxiety measured before boarding should predict the dog's cortisol and heart rate at the moment of separation, over and above the animal's baseline temperament.
- Interventions directed at the owner , and not only at the dog, should reduce the animal's stress markers: a dyad is treatable from either of its two poles.
- The dog–owner cortisol synchrony documented over the long term (Sundman et al., 2019) should intensify acutely around the separation event.
- Counterintuitively, dogs with closer bonds or longer cohabitation should show greater acute transmission, not lesser, consistent with the modulation of emotional contagion by duration of ownership.
- The owner's scent within the crate — commonly recommended as comfort — should have an ambivalent effect: a safety signal if the owner was calm when impregnating it, and a potential vehicle for stress chemosignals if they were not.

Under this model, the most frequently cited gap in the field — the absence of physiological measurements inside a commercial hold — ceases to be a deficiency and becomes an agenda: the model specifies what to measure (dyadic markers, not only the animal's), when (in the separation window, not in cruise) and in whom (in both poles, not one). It also explains why such measurement does not and will not readily exist: no one funds an instrumented cargo flight, so that knowledge in this field has an observational ceiling. Acknowledging this does not weaken the model; it delimits its scope and converts inference by convergence into the legitimate instrument — not a substitute — for the experiment that will not be conducted. Within that agenda, the model flags two populations currently without evidence of their own as priorities: the cat, whose response in real flight is almost non-existent (Jahn and DePorter, 2023), and the dog with cardiac disease, never monitored during air transport.

9. Conclusion

The risk perceived by the owner and the risk observed in official records are misaligned: official data place the physical risk of hold transport as low. The determinant of distress is neither the pressurized compartment nor the dog considered in isolation, but the dyad: the owner is the source of the stress — generating symptoms of their own and transmitting them to a dog that evolution turned into their emotional agent — while separation and flight stressors act as triggers upon that coupled system. The contribution of this note is interdisciplinary: it integrates the veterinary physiology of stress with the human psychology of its origin, upon a neuroendocrine substrate common to both species and evidence of transmission measured in first-quartile sources. The framework is explicitly observational and delimited — it describes and bounds, it does not demonstrate a mechanism inside the hold — and remains, in every case, subsidiary to individual veterinary assessment, which translates general evidence into a decision pertinent to the individual animal.

Author contributions and delimitation of competence

J. Y. Camacho García (Doctor of Veterinary Medicine and Animal Science, CMVP 12434) is responsible for the veterinary clinical and physiological content of this work: transport stress physiology, risk stratification and management considerations.

C. E. Ravello Joo participates in the capacity of a psychology student . His contribution is circumscribed to the analysis of human–animal behaviour and to the integration of the human component within the proposed dyadic model. This contribution does not constitute the practice of clinical psychology, does not formulate diagnosis or clinical judgement regarding persons, and must not be interpreted as psychological advice. Statements concerning human behaviour are presented on a descriptive and observational plane, and none of them substitutes assessment by a professional licensed in the corresponding field.

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